(a) One shared latent world model

observed state

current o_t | future o_{t+h} | goal o_g

future frames: training only

Image tokens

spatial

+ NULL

Latent–action tokens

State history

z_{t-1} z_t

Physical actions

a_{t-1} a_t

z a z a

causal

Shared Transformer Fθ

Shared readout

same trainable weights

read

Encoded states

z_t z_{t+h} z_g

predict

Latent rollout

$\hat{z}_{t+1}$ … $\hat{z}_{t+H}$

Separate spatial / causal tokenization and attention masks

autoregressive latent update · no future images

(b) Goal-conditioned MPC

Encode observation and goal with Fθ

z_t z_g

CEM / MPC

Sample candidate action sequences

Latent rollout · reuse Fθ

$J = \|\hat{z}_{t+H} - z_g\|^2 / d$

Update elite action distributions

STOP if $\|z_t - z_g\|^2 / d \le \varepsilon$

using the current observed state

first action

Environment

new observation

Goal enters the cost, not dynamics.

(c) Joint training

Encoded future targets

z_{t+h}

Predicted future states

$\hat{z}_{t+h}$

Multi-step MSE

Encoded states z

SIGReg(z)

L = MSE + λ SIGReg(z)

Both branches receive gradients

Shared readout weights; modality-specific BatchNorm statistics.